

ISBN: 978-81-960780-0-3

Biodiversity of Western Ghats



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Vimala College Publications
Thrissur, Kerala, India 680 009

BIODIVERSITY OF WESTERN GHATS

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ISBN: 978-81-960780-0-3

Printed at Bhavana Print Factory, Thrissur

Price: 130/-

Vimala College Publications

Vimala College (Autonomous), Thrissur, Kerala, 680009, India

Tel: 0487-2332080

Email: Vimalacollegepublications@gmail.com

Web: www.vimalacollege.edu.in



PREFACE

India's West Coast is bordered by the Western Ghats, which occupy a region of 180000 square kilometers. A mega biodiversity zone and biodiversity hotspot with a wide variety of flora, fauna, and landscape are known as the "Western Ghats". The Western Ghats have exceptionally high levels of endemism and plant and animal variety, as well as a significant number of species that are threatened with extinction internationally. The natural ecosystems of this hotspot are in danger due to the adverse effects of human activity on the environment, so immediate conservation measures are required, particularly the expansion of the protected area network. The chance to profit from biodiversity to the fullest possible will be lost forever if no steps are done to preserve it. The area is also home to several nationally notable animal sanctuaries, tiger reserves, and national parks, as well as a good collection of large mammals. Find numerous medicinal plants as well as significant genetic resources, such as wild relatives of cereals, fruits, and spices available in the Western Ghats. The Western Ghats are home to a variety of social, religious, and linguistic groups in addition to a rich biodiversity. This book primarily focuses on the diversity of floral and specific faunal elements, especially the bird varieties.

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THE NILGIRIS: A BIOGEOGRAPHIC HOTSPOT FOR THE ORCHID DIVERSITY

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1.1. INTRODUCTION

Orchidaceae, one of the diversified group among the angiosperms is distributed with ca. 28,000 taxa in tropics, sub-tropics and temperate regions of the world. They are known to occupy a variety of niches with habitats ranging from terrestrial to epiphytic and even underground (Christenhusz & Byng 2016, Adit & al. 2021). While our country, India experience the Tropical wet and dry climate with varied topographic factors thus flourishing the terrain into ten biogeographic regions. These influence the country for rich biodiversity with 11.4% of the total species from earth. Among these, Indian orchids is recorded with 1268 taxa of 155 genera with 307 endemics (Singh & al. 2020). Also, the orchids are rare with limited population due to high environmental threats viz. habitat loss, land use change, road expansion, and pollution. Therefore, the Orchidaceae family is recognised as Threatened Group of flowering plants.

1.2. The Nilgiris of the Western Ghats

The Western Ghats is one of the four Biodiversity Hotspots in India, running parallel to the Arabian coast from 8°N to 21°N latitude covering 1,60,000 km². These undulating mountains are distributed in the States of Gujarat, Maharashtra, Goa, Karnataka, Kerala, and Tamil Nadu. Amidst a Palghat gap with 30 km wide interrupt

the continuous chain of hills between Nilgiris and Anamalais. Due to high floristic richness and high concentration of endemic plants Western Ghats is divided into two sub-divisions as northern Western Ghats [includes Konkan and Raigad-Khandala Hills; and Shimoga-Chikmagalur-Dakshina Kannada of Karnataka] and southern Western Ghats [includes Malabar coast and Northern Kodagu; and Nilgiri-Wayanad-Silent Valley plateau and hills; Palni hills; Anamalai and High range hills; Agasthyamalai-Tirunelveli and Mahendragiri hills].

Besides various regions in the Western Ghats, **The Nilgiris** is endowed with unique feature of topography and climate. Based on this, the area is divided into The Nilgiri Plateau (central part), The Sigur Plateau (lying at the foot hills on the northern side of the former), The Nilgiri-Wayanad (lying on the western side of the above two plateau) and the outer slopes facing the plains (Sharma *et al.* 1977, Joseph 1987). The Nilgiris District lies in the state of Tamil Nadu positioned at 11°12 N - 11°37 N Latitude and 76°30 E - 76°55 E Longitude with the altitude between 300-2,637 m and covering geographical area about 2,565 km²; very dense forest, moderate dense forest and open forest with 460.67 km², 629.51 km² and 641.20 km² respectively. It is bounded on the north by Karnataka state, south-west by Kerala and; east and south-east by Coimbatore District, Tamil Nadu. Almost all important forest types of south India are represented here viz. Southern Tropical Thorn Forests, Southern Tropical Dry Deciduous Forests, Southern Tropical Moist Deciduous Forests, Tropical Semi-Evergreen Forests, Tropical Wet Evergreen Forests, Sub-tropical Hill Savannah and Southern Montane Wet Temperate Forests (Champion and Seth 1968) are distributed from the elevation 300-2,637 m.

1.3. The Nilgiris favours orchid diversity

The Nilgiris is the treasure hunt for the Naturalist; diversified with rich flora and fauna. For this reason, The Protected Areas was formed in Nilgiri and declared as Nilgiri Biosphere Reserve in 1986 covering an area about 5,520 km², Mukurthi

National Park with 78.46 km² in 1982 and Mudumalai Tiger Reserve with 321 km² in 2007. The Nilgiris is recognised as one of the highest rich biodiversity regions of the Western Ghats; and favourable habitation of orchid diversity with 138 species of 51 genera (**Table 1 & Plates 1-7**). The endemism accounts with 55 species to the study area (40 species endemic to the Western Ghats, 07 species endemic to the Eastern and Western Ghats, 03 species endemic to Tamil Nadu, and 05 species distributed throughout India). This is due to a distinct geographical location, topography, wide range of altitudinal variations from 700 – 2637 m, marked climate regimes, significant diurnal range of solar radiation and temperature (Singh & al. 2015). These factors determine the orchid diversity on each zonation of vegetation; where the species richness increases along the altitude and in mid elevation peak the diversity is high having an empirical support for the species (Rahbek 1995).

1.4. Risk factors for orchid survival

Besides its unique distribution pattern, the orchids face high risk for its survival from habitat loss, torrential rainfall, landslides, encroachment, over exploitation for medicinal purposes, road extension, and pollution etc., Therefore an urgent need for raising the awareness on orchid should be dispersed among the public, school and college students, and local people. This could be facilitated through participatory approach. The community participation (local people) helps to recognise the importance of orchids and wealth of their own forests. It benefits with conservation of orchids and biodiversity of an area which contributes towards the Sustainable Development Goals (SDG's).

ACKNOWLEDGMENT

The authors are grateful DST-NRDMS, Ministry of Science & Technology for funding the project. Also extend our thanks to The Principal Chief Conservator

of Forest, Chennai, for granting permission to carry out the research work in the forests.

CONFLICT OF INTEREST

The authors declare no conflict of interest.

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Table 1: Enumeration of Orchids in the Nilgiris District, Tamil Nadu

Sl. No	Name of the species	*Habit	Local distribution of the Nilgiris	Distribution
1.	<i>Acampe praemorsa</i> (Roxb.) Blatt. & McCann	E	Nellithorai	Andhra Pradesh, Chhattisgarh, Dadara & Nagar Haveli, Goa, Gujarat, Jharkhand, Karnataka, Kerala, Madhya Pradesh, Maharashtra, Odisha, Rajasthan and Tamil Nadu. Sri Lanka, Nepal, Myanmar and Seychelles.
2.	<i>Acanthophippium bicolor</i> Lindl.	T	Nilgiris	Karnataka, Kerala, Odisha and Tamil Nadu. Sri Lanka.
3.	<i>Aerides crispa</i> Lindl.	E	Avalanche, Parkside, Singara, Kodanad, Choladi river-Cherambadi, Rockwood	Goa, Gujarat, Karnataka, Kerala, Maharashtra and Tamil Nadu. Sri Lanka, Myanmar and Bangladesh.
4.	<i>Aerides ringens</i> (Lindl.) C.E.C.Fisch.	E	Avalanche, Naduvattam, Glenmorgan, Gulalur-Cherambadi, Kunnakobai, Kundha, Lakkad, Karia shola, Kodanad, Curzon valley	Andhra Pradesh, Goa, Gujarat, Karnataka, Kerala, Odisha and Tamil Nadu. Sri Lanka.
5.	<i>Anoectochilus elatus</i> Lindl.	T	Davala, Lambs rock-Coonoor	Karnataka, Kerala and Tamil Nadu. Sri Lanka.

6.	<i>Aphyllorchis montana</i> Rchb.f.	T	Nilgiris	Andhra Pradesh, Arunachal Pradesh, Karnataka, Kerala, Manipur, Meghalaya, Mizoram, Nagaland, Sikkim, Tamil Nadu and West Bengal. Sri Lanka, China, Japan, Indonesia, Thailand, Cambodia, Vietnam, Malaysia and Philippines.
7.	<i>Brachycorythis iantha</i> (Wight) Summerh.	T	Pykara, Porthimund, Mudimund, Naduvattam-Devarshola	Kerala and Tamil Nadu. Endemic to the Western Ghats.
8.	<i>Bulbophyllum acutiflorum</i> A.Rich.	E	Nilgris	Karnataka, Kerala and Tamil Nadu. Endemic to the Western Ghats.
9.	<i>Bulbophyllum elegantulum</i> (Rolfe) J.J.Sm.	E	Kudini	Karnataka, Kerala and Tamil Nadu. Endemic to the Western Ghats.
10.	<i>Bulbophyllum fimbriatum</i> (Lindl.) Rchb.f.	E	Gudalur, Puliyanbara	Goa, Karnataka, Kerala, Maharashtra and Tamil Nadu. Endemic to the Western Ghats.
11.	<i>Bulbophyllum fischeri</i> Seidenf.	E	Naduvattam, Katteri falls	Karnataka, Kerala and Tamil Nadu. Sri Lanka, Cambodia, China and Laos.
12.	<i>Bulbophyllum fuscopurpureum</i> Wight	E	Naduvattam	Karnataka, Kerala and Tamil Nadu. Endemic to the Western Ghats.
13.	<i>Bulbophyllum kaitiense</i> Rchb.f.	E	Terras valley	Karnataka, Kerala and Tamil Nadu. Endemic to the Western Ghats.
14.	<i>Bulbophyllum mysorensense</i> (Rolfe) J.J.Sm.	E	Nilgiris	Karnataka, Kerala and Tamil Nadu. Endemic to the Western Ghats.

15.	<i>Bulbophyllum nodosum</i> (Rolfe) J.J.Sm.	E	Nilgiris	Arunachal Pradesh and Tamil Nadu. Endemic to India.
16.	<i>Bulbophyllum proudlockii</i> (King & Pantl.) J.J.Sm.	E	Nilgiris	Karnataka and Tamil Nadu. Endemic to the Western Ghats.
17.	<i>Bulbophyllum sterile</i> (Lam.) Suresh	E	Nadugani, Ooty, Coonoor	Andhra Pradesh, Goa, Karnataka, Kerala, Maharashtra and Tamil Nadu. Nepal, Bangladesh and Myanmar.
18.	<i>Bulbophyllum tremulum</i> Wight	E	Naduvattum	Karnataka, Kerala and Tamil Nadu. Endemic to the Western Ghats.
19.	<i>Calanthe sylvatica</i> (Thouars) Lindl.	T	Nilgiris	Assam, Karnataka, Kerala, Maharashtra, Mizoram, Tamil Nadu and West Bengal. Sri Lanka, Tropical and Subtropical Asia and Africa.
20.	<i>Calanthe triplicata</i> (Willemet) Ames	T	Kotagiri, Doddabetta, Coonoor, Avalanche	Andaman & Nicobar Islands, Arunachal Pradesh, Kerala, Manipur, Meghalaya, Mizoram, Nagaland, Odisha, Tamil Nadu and West Bengal. Sri Lanka, China, Thailand, Japan, Malaysia, Pacific Islands and Australia.
21.	<i>Cheirostylis flabellata</i> (A.Rich.) Wight	E	Coonoor, Pakasura Kalhatti	Karnataka, Kerala, Maharashtra and Tamil Nadu. Sri Lanka, Bhutan and Myanmar.
22.	<i>Cleisostomopsis filiformis</i> (Rchb.f.) R.Rice	E	Sispara, Doddabetta, Pykara, T.R. Bazaar,	Kerala and Tamil Nadu. Sri Lanka.

			Carriott shola, Lakkadi, Korakundha	
23.	<i>Cleisostomopsis roseus</i> (Wight) R.Rice	E	Naduvattum	Karnataka, Kerala and Tamil Nadu. Endemic to the Western Ghats.
24.	<i>Coelogyne breviscapa</i> Lindl.	E	Singara	Karnataka, Kerala and Tamil Nadu. Sri Lanka.
25.	<i>Coelogyne mossiae</i> Rolfe	E	Nilgris	Kerala and Tamil Nadu. Endemic to the Western Ghats.
26.	<i>Coelogyne nervosa</i> A.Rich.	E	Kotagiri, Kulkutty, Kodanad, Naduvattam, Pakkasura hills, Pykara	Karnataka, Kerala and Tamil Nadu. Endemic to the Western Ghats.
27.	<i>Coelogyne odoratissima</i> Lindl.	E	Avalanche, Naduvattam, Doddabetta, Pykara, Nilgiri peak, T.R. Bazaar shola, Lakkadi, Mudimundu, Bangihalla, Kollimund	Karnataka, Kerala and Tamil Nadu. Sri Lanka.
28.	<i>Coelogyne uniflora</i> Lindl.	E	Nilgiris	Assam and Tamil Nadu. Cambodia, China South-Central, East Himalaya, India, Laos, Malaya, Myanmar, Nepal, Thailand and Vietnam.
29.	<i>Corymborkis veratrifolia</i> (Reinw.) Blume	T	Nilgiris	Andaman & Nicobar Islands, Arunachal Pradesh, Assam, Kerala, Manipur, Meghalaya, Mizoram, Nagaland, Sikkim, Tamil Nadu and

				West Bengal. Bangladesh, Sri Lanka, China, Myanmar, Indonesia, Japan, Malaysia, Thailand, Indo-China, Australia and Pacific Islands.
30.	<i>Cottonia peduncularis</i> (Lindl.) Rchb.f.	E	Gudalur-Killur	Goa, Karnataka, Kerala, Maharashtra, Odisha and Tamil Nadu. Sri Lanka.
31.	<i>Crepidium acuminatum</i> (D.Don) Szlach.	T	Naduvattam, Cherambadi, Kottamalai	Assam, Manipur, Meghalaya, Mizoram, Nagaland, Arunachal Pradesh, Sikkim, West Bengal, Himachal Pradesh, Uttarakhand, Karnataka, Kerala and Tamil Nadu. Bangladesh, Bhutan, Nepal, China, Myanmar, Indonesia, Philippines, Thailand, Indo-China and Australia.
32.	<i>Crepidium densiflorum</i> (A.Rich.) Sushil K.Singh, Agrawala & Jalal	T	Naduvattam, Devarshola, Bison Swamp, Bimaka shola, Pykara	Karnataka, Kerala, Odisha and Tamil Nadu. Sri Lanka.
33.	<i>Crepidium intermedium</i> (A.Rich.) Sushil K.Singh, Agrawala & Jalal	T	Naduvattam, Mulakadu- Ouchterlony valley, Pykara	Karnataka, Kerala and Tamil Nadu. Endemic to the Western Ghats.
34.	<i>Crepidium versicolor</i> (Lindl.) Sushil K.Singh, Agrawala & Jalal	T	Mulakadu- Ouchterlony valley, Naduvattam	Andhra Pradesh, Goa, Karnataka, Kerala, Maharashtra, Odisha and Tamil Nadu. Sri Lanka.

35.	<i>Cylindrolobus pauciflorus</i> (Wight) Schuit., Y.P.Ng & H.A.Pedersen	E	Coonoor, Kadanad, Kattery falls	Karnataka, Kerala and Tamil Nadu. Endemic to the Western Ghats.
36.	<i>Cymbidium aloifolium</i> (L.) Sw.	E	Coonoor	Throughout India. Sri Lanka, China, Myanmar, Bangladesh, Laos, Cambodia, Vietnam, Malaysia and Indonesia.
37.	<i>Dendrobium barbatulum</i> Lindl.	E	Nadugani	Goa, Gujarat, Karnataka, Kerala, Maharashtra and Tamil Nadu. Endemic to the Western Ghats.
38.	<i>Dendrobium herbaceum</i> Lindl.	E	Gudalur	Andhra Pradesh, Bihar, Chhattisgarh, Goa, Jharkhand, Karnataka, Kerala, Madhya Pradesh, Maharashtra, Mizoram, Odisha, Tamil Nadu and West Bengal. Bangladesh.
39.	<i>Dendrobium heyneanum</i> Lindl.	E	Naduvattam	Karnataka, Kerala and Tamil Nadu. Endemic to the Western Ghats.
40.	<i>Dendrobium jerdonianum</i> Wight	E	Naduvattam	Gujarat, Karnataka, Kerala, Maharashtra and Tamil Nadu. Sri Lanka.
41.	<i>Dendrobium macrostachyum</i> Lindl.	E	Devala, Maasikadu, Cherambadi-Erumadu, Kuzhivayal, Kolikara	Arunachal Pradesh, Goa, Jharkhand, Karnataka, Kerala, Maharashtra, Odisha, Tamil Nadu, Uttarakhand and West Bengal. Sri Lanka, Samoa Island, Sumatra, Java, Myanmar, Thailand, Malaysia, Bali and Australia.

42.	<i>Dendrobium microbulbon</i> A.Rich.	E	Naduvattam	Goa, Gujarat, Karnataka, Kerala, Maharashtra and Tamil Nadu. Endemic to the Western Ghats.
43.	<i>Dendrobium nanum</i> Hook.f.	E	Nelakota-Davala Road	Goa, Gujarat, Karnataka, Kerala, Maharashtra and Tamil Nadu. Endemic to the Eastern and Western Ghats.
44.	<i>Dendrobium nodosum</i> Dalzell	E	Gudalur	Goa, Karnataka, Kerala, Maharashtra and Tamil Nadu. Endemic to the Western Ghats.
45.	<i>Dendrobium wightii</i> A.D.Hawkes & A.H.Heller	E	Naduvattam	Karnataka, Kerala and Tamil Nadu. Endemic to the Western Ghats.
46.	<i>Diplocentrum recurvum</i> Lindl.	E	Kambatti	Andhra Pradesh, Karnataka, Kerala, Maharashtra and Tamil Nadu. Sri Lanka.
47.	<i>Disperis neilgherrensis</i> Wight	T	Naduvattam, T.R. Bazar shola, Gudalur, Ouchterlony valley	Karnataka, Kerala, Meghalaya, Odisha and Tamil Nadu. Sri Lanka, Nepal, Malaysia, Java, New Caledonia, Australia and Africa.
48.	<i>Epipactis veratrifolia</i> Boiss. & Hohen.	T	Devarshola	Sikkim, West Bengal, Himachal Pradesh, Jammu & Kashmir, Uttarakhand, Punjab, Rajasthan and Tamil Nadu. Afghanistan, Nepal, Pakistan, China, Myanmar, South West Asia, Russia and Africa.
49.	<i>Epipogium roseum</i> (D.Don) Lindl.	T	Ouchterlony valley	Arunachal Pradesh, Himachal Pradesh,

				Jammu and Kashmir, Karnataka, Kerala, Maharashtra, Meghalaya, Mizoram, Sikkim, Tamil Nadu, Uttarakhand and West Bengal. Sri Lanka, Nepal, Malaysia, Java, New Caledonia, Australia and Africa.
50.	<i>Eria albiflora</i> Rolfe	E	T.R. Bazzar	Karnataka, Kerala and Tamil Nadu. Endemic to the Western Ghats.
51.	<i>Eulophia cernua</i> (Willd.) M.W.Chase, Kumar & Schuit.	T	Gudalur Ghat	Throughout India. Sri Lanka, Nepal, China, Myanmar, Bangladesh, Thailand, Malaysia, Philippines, Nanseishoto, Australia and Fiji.
52.	<i>Eulophia epidendraea</i> (J.Koenig ex Retz.) C.E.C.Fisch.	T	Kallar	Andaman & Nicobar Islands, Andhra Pradesh, Chhattisgarh, Karnataka, Kerala, Maharashtra, Odisha, Tamil Nadu, Uttarakhand and West Bengal. Sri Lanka and Bangladesh.
53.	<i>Eulophia nuda</i> Lindl.	T	Carcoor Ghat, Naduvattam, Coonoor, Kundha	Throughout India. Sri Lanka, Nepal, China, Myanmar, Bangladesh, Thailand, Malaysia, Philippines and Pacific Island.
54.	<i>Eulophia pulchra</i> (Thouars) Lindl.	T	Manar river side – Parali	Karnataka, Kerala, Maharashtra and Tamil Nadu. Sri Lanka, Taiwan, Laos, Cambodia, Vietnam, Thailand, Malaysia,

				Philippines, Madagascar, Africa and Australia.
55.	<i>Gastrochilus calceolaris</i> (Buch. - Ham. ex Sm.) D.Don	E	Kanavari, Kilkotagiri, Kodanad, Ermadu, Cherambadi	Andaman & Nicobar Islands, Andhra Pradesh, Arunachal Pradesh, Assam, Himachal Pradesh, Manipur, Meghalaya, Mizoram, Nagaland, Sikkim, Tamil Nadu, Tripura, Uttarakhand and West Bengal.
56.	<i>Goodyera procera</i> (Ker Gawl.) Hook.	T	Purliar, Devala- Pandalur Forest	Throughout India. Sri Lanka, Bangladesh, Bhutan, Cambodia, China, East Timor, Indonesia, Japan, Laos, Malaysia, Myanmar, Nepal, Philippines, Thailand, Taiwan and Vietnam.
57.	<i>Habenaria barnesii</i> Summerh. ex C.E.C.Fisch.	T	Mudimund	Kerala and Tamil Nadu. Endemic to the Western Ghats.
58.	<i>Habenaria brachyphylla</i> (Lindl.) Aitch.	T	Naduvattam, Gudalur, Ebanad-Sirur, Devarshola	Karnataka, Kerala, Maharashtra, Odisha and Tamil Nadu. Endemic to the Eastern and Western Ghats.
59.	<i>Habenaria cephalotes</i> Lindl.	T	Doddabetta, Pykara, Naduvattam	Karnataka, Kerala, Maharashtra and Tamil Nadu. Endemic to the Western Ghats.
60.	<i>Habenaria denticulata</i> Rchb.f.	T	Nilgiris	Tamil Nadu. Endemic to Tamil Nadu in Western Ghats.
61.	<i>Habenaria digitata</i> Lindl.	T	Naduvattam, Avalanche	Andhra Pradesh, Bihar, Chhattisgarh, Goa, Himachal Pradesh,

				Jharkhand, Karnataka, Kerala, Maharashtra, Meghalaya, Odisha, Sikkim, Tamil Nadu, Uttarakhand and West Bengal. Bangladesh, Myanmar, Nepal, Pakistan, Laos and Cambodia.
62.	<i>Habenaria elliptica</i> Wight	T	Nilgiris	Karnataka, Kerala and Tamil Nadu. Endemic to the Western Ghats.
63.	<i>Habenaria elwesii</i> Hook.f.	T	Devala	Goa, Karnataka, Kerala and Tamil Nadu. Endemic to the Western Ghats.
64.	<i>Habenaria furcifer</i> a Lindl.	T	Mudumalai, Benne regions	Throughout India. Bhutan, Nepal, Pakistan, Myanmar, Bangladesh, China, Laos and Thailand.
65.	<i>Habenaria heyneana</i> Lindl.	T	Ooty, Naduvattam, Pykara, Mudimund, Kariashola	Andhra Pradesh, Goa, Karnataka, Kerala, Maharashtra and Tamil Nadu. Endemic to the Eastern and Western Ghats.
66.	<i>Habenaria hollandiana</i> Santapau	T	Nilgiris	Andhra Pradesh, Karnataka, Maharashtra, Tamil Nadu and Jharkhand. Endemic to India.
67.	<i>Habenaria longicorniculata</i> J. Graham	T	Devala, Coonoor, Naduvattam, Avalanche, Nadugani	Andhra Pradesh, Chhattisgarh, Jharkhand, Odisha, Goa, Gujarat, Karnataka, Kerala, Madhya Pradesh, Maharashtra, Rajasthan and Tamil Nadu. Sri Lanka.

68.	<i>Habenaria longicornu</i> Lindl.	T	Coonoor, Pakkasuramalai, Ebanad, Kodanad	Madhya Pradesh, Karnataka, Kerala and Tamil Nadu. Endemic to India.
69.	<i>Habenaria multicaudata</i> Sedgw.	T	Manjur	Goa, Karnataka, Kerala, Maharashtra and Tamil Nadu. Endemic to the Western Ghats.
70.	<i>Habenaria plantaginea</i> Lindl.	T	Mudumalai, Carcoor Ghat, Moyar	Andhra Pradesh, Chhattisgarh, Goa, Haryana, Jharkhand, Karnataka, Kerala, Madhya Pradesh, Maharashtra, Odisha, Punjab, Uttarakhand, Uttar Pradesh and West Bengal. Sri Lanka, Myanmar, Bangladesh and Nepal.
71.	<i>Habenaria polyodon</i> Hook.f.	T	Manjur	Tamil Nadu. Endemic to Tamil Nadu.
72.	<i>Habenaria rariflora</i> A. Rich.	T	Coonoor-Lambs rock, Pykara, Pakkasuramalai Anakatti area, Kodanad	Andhra Pradesh, Karnataka, Kerala, Maharashtra, and Tamil Nadu. Endemic to the Eastern and Western Ghats.
73.	<i>Habenaria richardiana</i> Wight	T	Doddabetta, Kodanad	Kerala, Maharashtra and Tamil Nadu. Endemic to the Western Ghats.
74.	<i>Liparis alata</i> A.Rich.	T	Nilgris	Tamil Nadu. Endemic to Tamil Nadu.
75.	<i>Liparis atropurpurea</i> Lindl.	T	Kundha, Mukurti, Pykara, Avalanche, Ouchterlony valley	Karnataka, Kerala and Tamil Nadu. Sri Lanka.

76.	<i>Liparis biloba</i> Wight	T	Kollimund	Karnataka and Tamil Nadu. Endemic to the Western Ghats.
77.	<i>Liparis cespitosa</i> (Lam.) Lindl.	T	Nilgiris	Arunachal Pradesh, Kerala, Manipur, Meghalaya, Mizoram, Nagaland, Tamil Nadu, Uttarakhand and West Bengal. Sri Lanka and Africa to Pacific Islands.
78.	<i>Liparis elliptica</i> Wight	T	Ittakalada	Andhra Pradesh, Arunachal Pradesh, Karnataka, Kerala, Manipur, Meghalaya, Odisha, Sikkim and Tamil Nadu. Sri Lanka, Nepal, Myanmar, China, Thailand, Taiwan, Indonesia, Philippines, Vietnam and Pacific Islands.
79.	<i>Liparis odorata</i> (Willd.) Lindl.	T	Cherankode	Andhra Pradesh, Arunachal Pradesh, Assam, Goa, Jharkhand, Karnataka, Kerala, Maharashtra, Manipur, Meghalaya, Mizoram, Nagaland, Odisha, Sikkim, Tamil Nadu, Uttarakhand and West Bengal. Sri Lanka, Nepal, China, Myanmar, Thailand, Japan and Java.
80.	<i>Liparis platyphylla</i> Ridl.	T	Benne forest	Karnataka and Tamil Nadu. Endemic to the Western Ghats.
81.	<i>Liparis resupinata</i> Ridl.	T	Nilgiris	Manipur, Meghalaya, Mizoram, Nagaland, Sikkim West Bengal,

				Uttarakahand and Tamil Nadu. Bangladesh, Bhutan, Nepal, China, Thailand and Indo-China.
82.	<i>Liparis viridiflora</i> (Blume) Lindl.		Meenmutti-Marva Halla Estate, Gudalur.	Andhra Pradesh, Assam, Karnataka, Kerala, Manipur, Meghalaya, Mizoram, Nagaland, Odisha, Sikkim, Tamil Nadu, Tripura, Uttarakhand and West Bengal. Sri Lanka, China, Nepal, Bhutan, Taiwan, Myanmar, Bangladesh, Malaysia, Philippines and Indonesia.
83.	<i>Liparis walkeriae</i> Graham	T	Nilgiris	Kerala and Tamil Nadu. Sri Lanka.
84.	<i>Liparis wightiana</i> Thwaites	T	Devarshola, Naduvattam	Karnataka, Kerala and Tamil Nadu. Sri Lanka, Thailand, Sumatra and Java.
85.	<i>Luisia tenuifolia</i> Blume	E	Lower Tiger hills, Bokkapuram, Bimaka shola, Deepdale, Mantharai shola	Goa, Karnataka, Kerala and Maharashtra. Sri Lanka.
86.	<i>Luisia tristis</i> (G.Forst.) Hook.f.	E	Deepdale, Gudalur-Sultan battery road	Throughout India. Sri Lanka, Nepal, China, Bangladesh, Myanmar, Bhutan, Malaysia and South East Asia.
87.	<i>Nervilia concolor</i> (Blume) Schltr.	T	Way to Thorapalli	Andaman & Nicobar Islands, Andhra Pradesh, Assam, Goa, Himachal Pradesh, Jharkhand, Karnataka,

				Kerala, Maharashtra, Manipur, Megalaya, Nagaland, Odisha, Rajasthan, Sikkim, Tamil Nadu, Uttarakhand, Uttar Pradesh and West Bengal. Thailand, Myanmar, Bangladesh, Bhutan, Malaysia, Java, Indonesia, Philippines, New Guinea and Australia.
88.	<i>Nervilia plicata</i> (Andrews) Schltr.	T	Northern Hay, Mudumalai	Throughout India. Bangladesh, Nepal, Pakistan, Myanmar, China, Thailand, Vietnam, Laos, Malaysia, Philippines and Australia.
89.	<i>Nervilia simplex</i> (Thouars) Schltr.	T	Gudalur	Andhra Pradesh, Arunachal Pradesh, Goa, Jharkhand, Kerala, Maharashtra, Manipur, Odisha, Sikkim, Tamil Nadu and Uttarakhand. Sri Lanka, Tropical and Subtropical Old World and Pacific Islands.
90.	<i>Oberonia arnottiana</i> Wight	E	Kottagiri ghat road	Kerala and Tamil Nadu. Endemic to the Western Ghats.
91.	<i>Oberonia brunoniana</i> Wight	E	Naduvattam, Pykara, Sispara, T.R. Bazzar shola, Bangihalla, Pykara	Andhra Pradesh, Goa, Karnataka, Kerala, Maharashtra and Tamil Nadu. Endemic to the Eastern and Western Ghats.
92.	<i>Oberonia chandrasekharanii</i> V.J. Nair,	E	Wilson plantation (Naduvattam)	Karnataka, Kerala and Tamil Nadu. Endemic to the Western Ghats.

	V.S.Ramach. & R. Ansari			
93.	<i>Oberonia ensiformis</i> (Sm.) Lindl.	E	Terras valley ghat road	Andaman & Nicobar Islands, Andhra Pradesh, Arunachal Pradesh, Karnataka, Kerala, Maharashtra, Manipur, Meghalaya, Mizoram, Nagaland, Odisha, Sikkim, Tamil Nadu, Uttarakhand and West Bengal. Nepal, China, Myanmar, Thailand, Laos and Vietnam.
94.	<i>Oberonia maxima</i> C.S.P.Parish ex Hook.f.	E	Mukurthi National Park	Arunachal Pradesh, Manipur, Meghalaya, Mizoram, Nagaland and Tamil Nadu. Myanmar and Thailand.
95.	<i>Oberonia mucronata</i> (D.Don) Ormerod & Seidenf.	E	Nilgiris	Andaman & Nicobar Islands, Andhra Pradesh, Arunachal Pradesh, Assam, Chhattisgarh, Goa, Gujarat, Karnataka, Kerala, Maharashtra, Manipur, Meghalaya, Mizoram, Nagaland, Odisha and Tamil Nadu. Nepal, Bangladesh, China, Myanmar, Thailand, Vietnam, Malaysia, Philippines and Indonesia.
96.	<i>Oberonia nayarii</i> R. Ansari & N.P. Balakr.	E	Naduvattam	Karnataka, Kerala and Tamil Nadu. Endemic to the Western Ghats.
97.	<i>Oberonia platycaulon</i> Wight	E	Naduvattam	Karnataka, Kerala and Tamil Nadu. Endemic to the Western Ghats.

98.	<i>Oberonia proudlockii</i> King & Pantl.	E	Nadugani	Karnataka, Kerala, Maharashtra, Odisha and Tamil Nadu. Endemic to the Eastern and Western Ghats.
99.	<i>Oberonia verticillata</i> Wight	E	Pykara, Naduvattam, Avalanche, Ebenad to Bikkampattim und, Kukkal Betta shola, Kodanad	Goa, Karnataka, Kerala, Maharashtra and Tamil Nadu. Endemic to the Western Ghats.
100	<i>Oberonia wightiana</i> Lindl.	E	Coonoor, Naduvattam road, Mudimund, Nilgiri peak R.F., Doddabetta, Avalanche, Kodanad, Bangihalla	Karnataka, Kerala and Tamil Nadu. Sri Lanka.
101	<i>Pachystoma pubescens</i> Blume	T	Pykara falls	Andaman & Nicobar Islands, Aruanchal Pradesh, Himachal Pradesh, Karnataka, Kerala, Madhya Pradesh, Maharashtra, Manipur, Meghalaya, Mizoram, Nagaland, Odisha, Punjab, Sikkim, Tamil Nadu, Uttarakhand, Uttar Pradesh and West Bengal. India, China, Bangladesh, Bhutan, Myanmar, Sumatra, Borneo, New Guinea, Philippines, Thailand, Malaysia and Java.

102	<i>Papilionanthe subulata</i> (Willd.) Garay	E	Naduvattam, Kattery falls, Matlada waterfalls, Kodanad, Kunnakkombai, Gudalur-Killur Forest	Arunachal Pradesh, Karnataka, Kerala and Tamil Nadu. Sri Lanka.
103	<i>Pecteilis gigantea</i> (Sm.) Raf.	T	Nadugani	Andhra Pradesh, Bihar, Chhattisgarh, Goa, Gujarat, Himachal Pradesh, Jammu & Kashmir, Jharkhand, Karnataka, Kerala, Madhya Pradesh, Maharashtra, Nagaland, Odisha and Tamil Nadu. Pakistan and Myanmar.
104	<i>Peristylus brachyphyllus</i> A.Rich.	T	Avalanche, Lakkadi	Karnataka and Tamil Nadu. Endemic to the Western Ghats.
105	<i>Peristylus goodyeroides</i> (D.Don) Lindl.	T	Gudalur, Devala, Benne, Mudumalai	Andhra Pradesh, Arunachal Pradesh, Assam, Bihar, Chhattisgarh, Himachal Pradesh, Jharkhand, Karnataka, Kerala, Madhya Pradesh, Maharashtra, Manipur, Meghalaya, Mizoram, Odisha, Sikkim, Tripura, Uttarakhand, Tamil Nadu and West Bengal. Sri Lanka, Bangladesh, Burma, Bhutan, China, Indonesia, Java, Malaysia, Nepal, New Guinea, Philipines, Sumatra, Thailand and Taiwan.

106	<i>Peristylus lawii</i> Wight	T	Dodagatti-Mudumalai	Andhra Pradesh, Bihar, Gujarat, Jharkhand, Karnataka, Kerala, Madhya Pradesh, Maharashtra, Odisha, Tamil Nadu and Uttarakhand. Nepal and Myanmar.
107	<i>Peristylus plantagineus</i> (Lind.) Lindl.	T	Gudalur	Andhra Pradesh, Chhattisgarh, Goa, Gujarat, Karnataka, Kerala, Madhya Pradesh, Maharashtra, Odisha, Tamil Nadu and West Bengal. Nepal and Sri Lanka.
108	<i>Peristylus richardianus</i> Wight	T	Avalanche	Arunachal Pradesh, Kerala, Maharashtra, Manipur, Meghalaya and Tamil Nadu. Nepal.
109	<i>Peristylus secundus</i> (Lindl.) Rathakr.	T	Nilgiris	Karnataka and Tamil Nadu. Endemic to the Western Ghats.
110	<i>Peristylus spiralis</i> A.Rich.	T	Avalanche, Mukurthi, Naduvattam	Karnataka, Kerala, Maharashtra and Tamil Nadu. Sri Lanka.
111	<i>Peristylus stocksii</i> (Hook.f.) Kraenzl.	T	Nilgiris	Bihar, Goa, Gujarat, Karnataka, Kerala, Madhya Pradesh, Maharashtra and Tamil Nadu. Endemic to India.
112	<i>Pholidota imbricata</i> Hook.	E	Kuzhivayal, Thallur forest, Nadugani, Ouchterlony valley, Ndukarna to Nedumballi-Waynaad	Throughout India. Sri Lanka, Tropical and Subtropical Asia and Pasific Islands.

113	<i>Pinalia mysorensis</i> (Lindl.) Kuntze	E	Kulhatty, Devarshola	Karnataka, Kerala, Maharashtra and Tamil Nadu. Endemic to the Western Ghats.
114	<i>Pinalia polystachya</i> (A.Rich.) Kuntze	E	Naduvattum, Kodanad shola	Karnataka, Kerala and Tamil Nadu. Endemic to the Western Ghats.
115	<i>Plectoglossa</i> <i>perrottetiana</i> (A.Rich.) K.Prasad	T	Avalanche	Karnataka, Kerala, Maharashtra and Tamil Nadu. Endemic to the Western Ghats.
116	<i>Podochilus falcatus</i> Lindl.	E	Nilgiris	Tamil Nadu. Sri Lanka.
117	<i>Polystachya</i> <i>concreta</i> (Jacq.) Garay & H.R.Sweet	E	Bimaka shola, Karyashola, Cherambadi- Choladi river	Andaman & Nicobar Islands, Andhra Pradesh, Arunachal Pradesh, Karnataka, Kerala, Manipur, Meghalaya, Nagaland, Odisha and Tamil Nadu. Sri Lanka, Myanmar, Indonesia, Malaysia, Philippines, Thailand, Indo-China, Africa and America.
118	<i>Porpax braccata</i> (Lindl.) Schuit., Y.P.Ng & H.A.Pedersen	E	Pykara, Coonoor, Kodanad	Goa, Karnataka, Kerala, Maharashtra, Tamil Nadu and Uttarakhand. Sri Lanka.
119	<i>Porpax exilis</i> (Hook.f.) Schuit., Y.P.Ng & H.A.Pedersen	E	Puliyampara, Gudalur	Goa, Karnataka, Kerala, Maharashtra and Tamil Nadu. Endemic to the Eastern and Western Ghats.
120	<i>Porpax filiformis</i> (Wight) Schuit., Y.P.Ng & H.A.Pedersen	E	Devala- Wyanad, Pandiar Dam- Nadugani	Goa, Gujarat, Karnataka, Kerala, Maharashtra and Tamil Nadu. Endemic to the Western Ghats.

121	<i>Porpax jerdoniana</i> (Wight) Rolfe	E	Nadugani, Gudalur	Andaman & Nicobar Islands, Goa, Karnataka, Kerala, Maharashtra and Tamil Nadu. Endemic to the Western Ghats and Andaman & Nicobar Island.
122	<i>Porpax nana</i> (A.Rich.) Schuit., Y.P.Ng & H.A.Pedersen	E	Kulhutti, Devarshola	Karnataka, Kerala and Tamil Nadu. Endemic to the Western Ghats
123	<i>Porpax reticulata</i> Lindl.	E	Cherankode	Goa, Karnataka, Kerala, Maharashtra and Tamil Nadu. Laos, Thailand and Vietnam.
124	<i>Rhynchostylis retusa</i> (L.) Blume	E	Sultan Battery, Thallur	Throughout India. Sri Lanka, Bhutan, Myanmar, Nepal, Bangladesh, China, Thailand, Laos, Cambodia, Vietnam, Malaysia, Philippines and Java.
125	<i>Robiquetia josephiana</i> Manilal & C.S.Kumar	E	Terras ghat road	Kerala and Tamil Nadu. Endemic to the Western Ghats.
126	<i>Satyrium nepalense</i> D.Don	T	Pykara, Thala Kundha, Naduvattam, Devarshola, Pakkasuramala i, Lakkadi	Andhra Pradesh, Arunachal Pradesh, Himachal Pradesh, Jammu & Kashmir, Karnataka, Kerala, Manipur, Meghalaya, Nagaland, Sikkim, Tamil Nadu, Uttarakhand and West Bengal. Sri Lanka, Nepal, Bhutan, Myanmar, China and Pakistan.
127	<i>Sirhookera latifolia</i> (Wight) Kutze	E	Puliyampara, Gudalur	Karnataka, Kerala and Tamil Nadu. Sri Lanka.

128	<i>Spiranthes sinensis</i> (Pers.) Ames	T	Coonoor, Pykara, Mukurti peak, Longwood shola, Kotagiri	Arunachal Pradesh, Assam, Himachal Pradesh, Jammu & Kashmir, Karnataka, Kerala, Maharashtra, Manipur, Meghalaya, Mizoram, Nagaland, Odisha, Punjab, Sikkim, Tripura, Tamil Nadu, Uttarakhand and West Bengal. Sri Lanka, Afghanistan, Bangladesh, China, Tibet, Thailand, Siberia, Japan, Java, Australia and New Zealand.
129	<i>Taeniophyllum</i> <i>alwisii</i> Lindl.	E	Naduvattam	Karnataka, Kerala and Tamil Nadu. Sri Lanka.
130	<i>Thrixspermum</i> <i>formosanum</i> (Hayata) Schltr.	E	Mukurthi National Park	Tamil Nadu. Taiwan and Vietnam.
131	<i>Thrixspermum</i> <i>musciflorum</i> A.S.Rao & J.Joseph	E	Nilgiris	Arunachal Pradesh, Assam, Kerala, Manipur, Meghalaya, Sikkim and Tamil Nadu. Endemic to India.
132	<i>Trichoglottis tenera</i> (Lindl.) Rchb.f.	E	Naduvattam- Gudalur forest, Carriot shola, Avalanche	Karnataka, Kerala and Tamil Nadu. Sri Lanka.
133	<i>Vanda spathulata</i> (L.) Spreng.	E	Upper Geddhai	Andhra Pradesh, Kerala and Tamil Nadu. Sri Lanka.
134	<i>Vanda tessellata</i> (Roxb.) Hook. ex D.Don	E	Doddabetta, Thengumarada , Parali forest	Throughout India. Sri Lanka, Myanmar, Nepal. Bangladesh and China.
135	<i>Vanda testacea</i> (Lindl.) Rchb.f.	E	Moyar bank, Bokkapuram, Avarihalla,	Throughout India. Sri Lanka, Nepal, Myanmar, Thailand,

			Anakatty, Iduhatti, Kunjapanai, Nellithorai slopes, Karkudi	Malaysia, Philippines and Indonesia.
136	<i>Vanda wightii</i> Rchb.f.	E	Nilgiris	Goa, Karnataka, Kerala and Tamil Nadu. Sri Lanka.
137	<i>Vanilla walkerae</i> Wight	T	Veerakkalai	Andhra Pradesh, Karnataka, Kerala and Tamil Nadu. Sri Lanka.
138	<i>Zeuxine longilabris</i> (Lindl.) Trimen	T	Kunjapanai	Arunachal Pradesh, Assam, Bihar, Goa, Karnataka, Kerala, Maharashtra, Odisha, Tamil Nadu, Tripura and West Bengal. Bangladesh, Cambodia, Malaysia, Nepal, Sri Lanka, Myanmar and Thailand.

*Habit: E – Epiphytes, T- Terrestrials

PLATE - 1



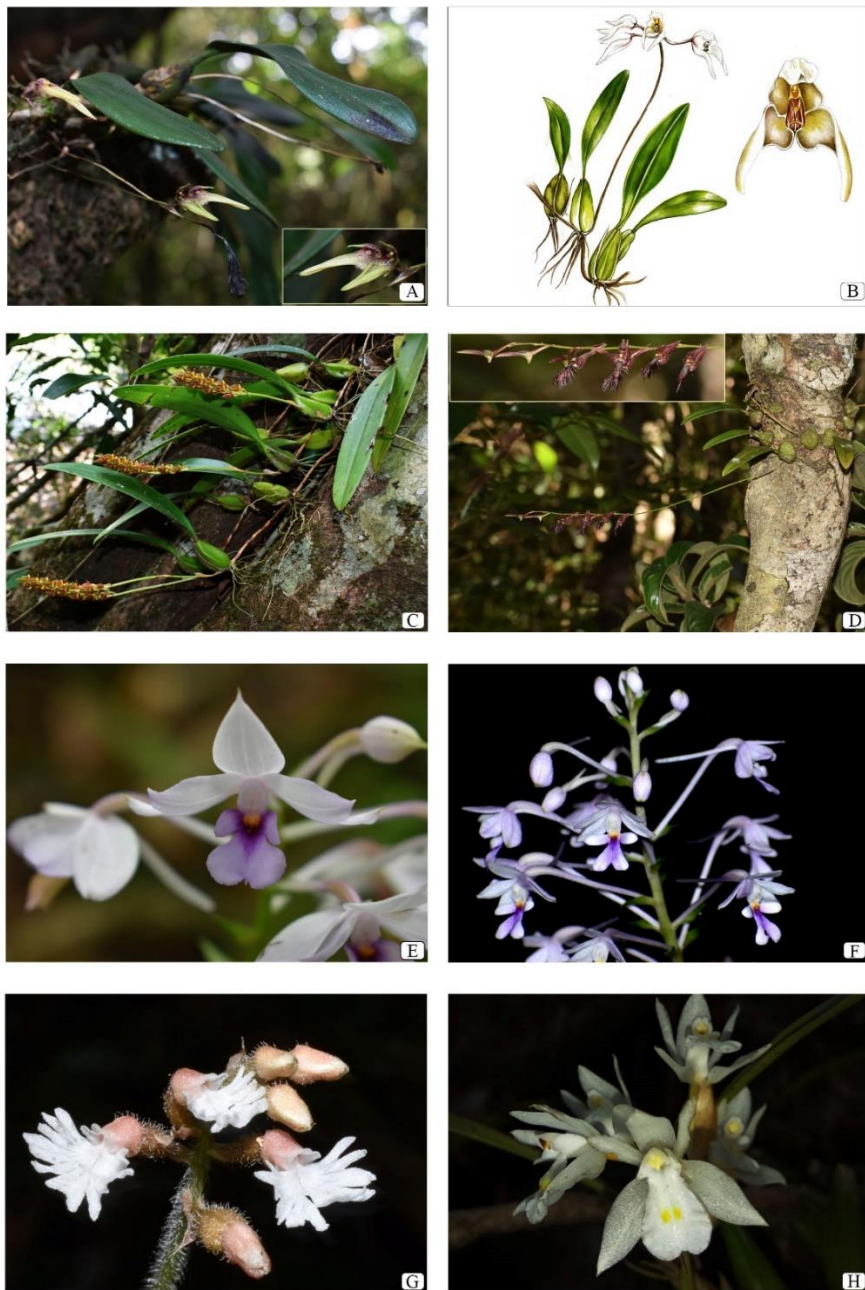
A - D. Orchid habitats, E - H. Orchid habits.

PLATE - 2



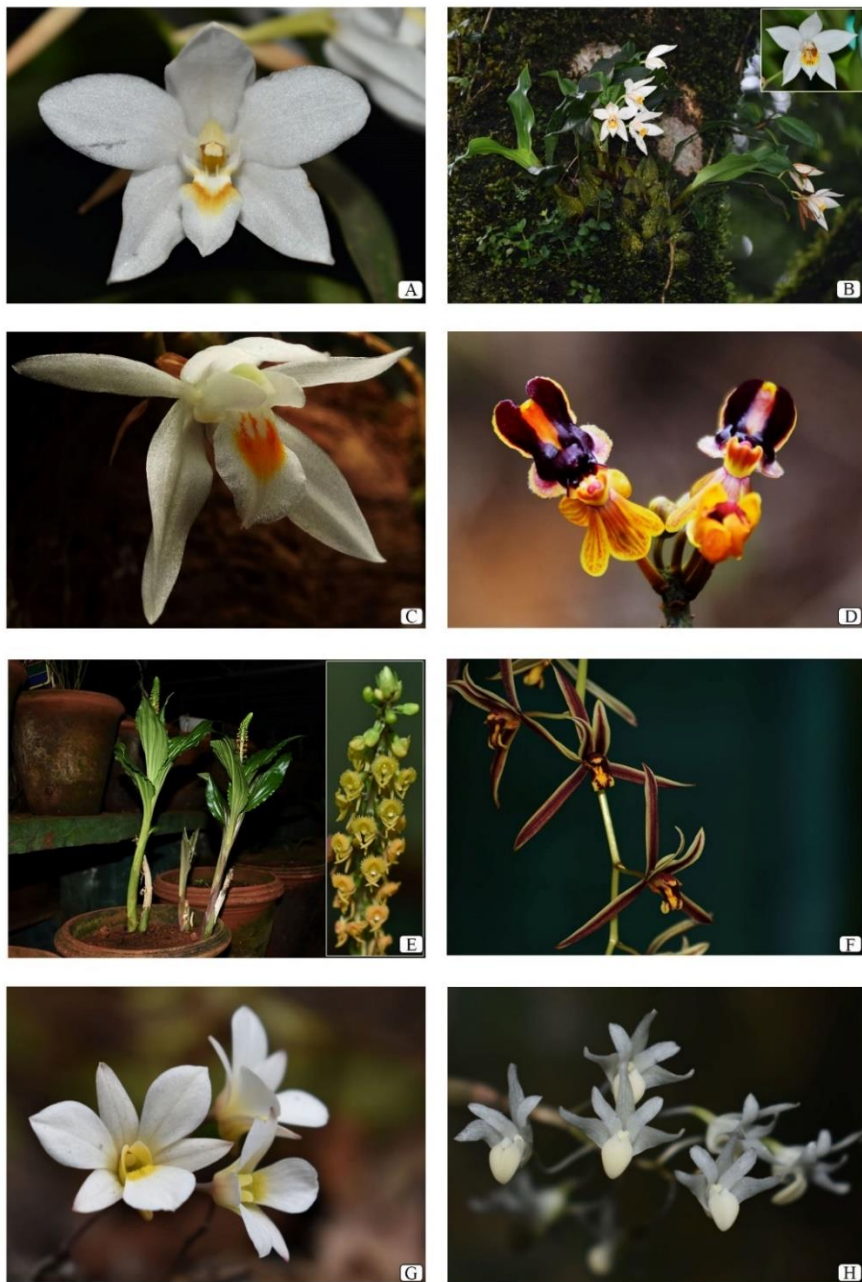
A. *Acampe praemorsa* (Roxb.) Blatt. & McCann, B. *Acanthophippium bicolor* Lindl., C. *Aerides crispa* Lindl., D. *Anoectochilus elatus* Lindl., E. *Aphyllorchis montana* Rehb.f., F. *Brachycorythis iantha* (Wight) Summerh., G. *Bulbophyllum acutiflorum* A.Rich., H. *Bulbophyllum fuscopurpureum* Wight

PLATE - 3



A. *Bulbophyllum kaitiense* Rchb.f., B. *Bulbophyllum mysorensense* (Rolfe) J.J.Sm., C. *Bulbophyllum sterile* (Lam.) Suresh, D. *Bulbophyllum tremulum* Wight, E. *Calanthe sylvatica* (Thouars) Lindl., F. *Calanthe triplicata* (Willemet) Ames, G. *Cheirostylis flabellata* (A.Rich.) Wight, H. *Coelogyne breviscapa* Lindl.

PLATE - 4



A. *Coelogyne mossiae* Rolfe, B. *Coelogyne nervosa* A.Rich., C. *Coelogyne odoratissima* Lindl., D. *Cottonia peduncularis* (Lindl.) Rchb.f., E. *Crepidium versicolor* (Lindl.) Sushil K.Singh, Agrawala & Jalal, F. *Cymbidium aloifolium* (L.) Sw., G. *Dendrobium barbatulum* Lindl., H. *Dendrobium herbaceum* Lindl.

PLATE - 5



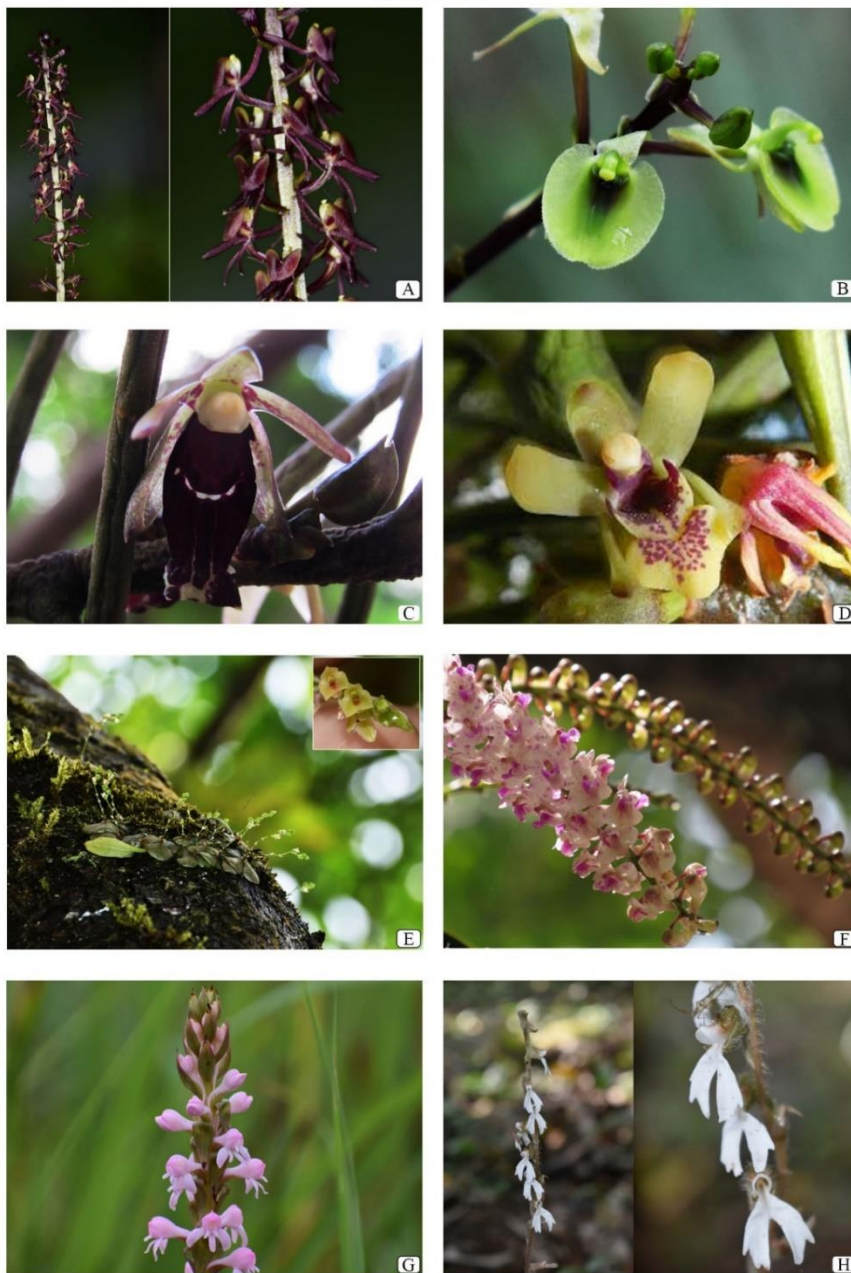
A. *Dendrobium heyneanum* Lindl., B. *Dendrobium macrostachyum* Lindl., C. *Dendrobium nanum* Hook.f., D. *Dendrobium nodosum* Dalzell, E. *Dendrobium wightii* A.D.Hawkes & A.H.Heller, F. *Diplocentrum recurvum* Lindl., G. *Disperis neilgherrensis* Wight, H. *Epipogium roseum* (D.Don) Lindl.

PLATE - 6



A. *Eria albiflora* Rolfe, B. *Eulophia epidendraea* (J.Koenig ex Retz.) C.E.C.Fisch., C. *Eulophia nuda* Lindl., D. *Eulophia pulchra* (Thouars) Lindl., E. *Goodyera procera* (Ker Gawl.) Hook., F. *Habenaria heyneana* Lindl., G. *Habenaria longicornu* Lindl., H. *Habenaria rariflora* A.Rich.

PLATE - 7



A. *Liparis atropurpurea* Lindl., B. *Liparis wightiana* Thwaites, C. *Luisia tenuifolia* Blume, D. *Luisia tristis* (G.Forst.) Hook.f., E. *Porpax exilis* (Hook.f.) Schuit., Y.P.Ng & H.A.Pedersen, F. *Rhynchosstylis retusa* (L.) Blume, G. *Satyrium nepalense* D.Don, H. *Zeuxine longilabris* (Lindl.) Trimen